
Alternative Architectures

Philipp Koehn

7 October 2025



Alternative Architectures



1

- We introduced one translation model
 - attentional seq2seq model
 - core organizing feature: recurrent neural networks
- Other core neural architectures
 - convolutional neural networks
 - attention
- But first: look at various components of neural architectures

components

Components of Neural Networks



- Neural networks originally inspired by the brain
 - a neuron receives signals from other neurons
 - if sufficiently activated, it sends signals
 - feed-forward layers are roughly based on this
- Computation graph
 - any function possible, as long as it is partially differentiable
 - not limited by appeals to biological validity
- *Deep learning* maybe a better name

Feed-Forward Layer



- Classic neural network component
- Given an input vector x , matrix multiplication M with adding a bias vector b

$$Mx + b$$

- Adding a non-linear activation function

$$y = \text{activation}(Mx + b)$$

- Notation

$$y = FF_{\text{activation}}(x) = a(Mx + b)$$

Feed-Forward Layer



- Historic neural network designs: several feed-forward layers
 - input layer
 - hidden layers
 - output layer
- Powerful tools for a wide range of machine learning problems
- Matrix multiplication also called **affine transforms**
 - appeals to its geometrical properties
 - straight lines in input still straight lines in output

Factored Decomposition



- One challenge: very large input and output vectors
- Number of parameters in matrix $M = |x| \times |y|$

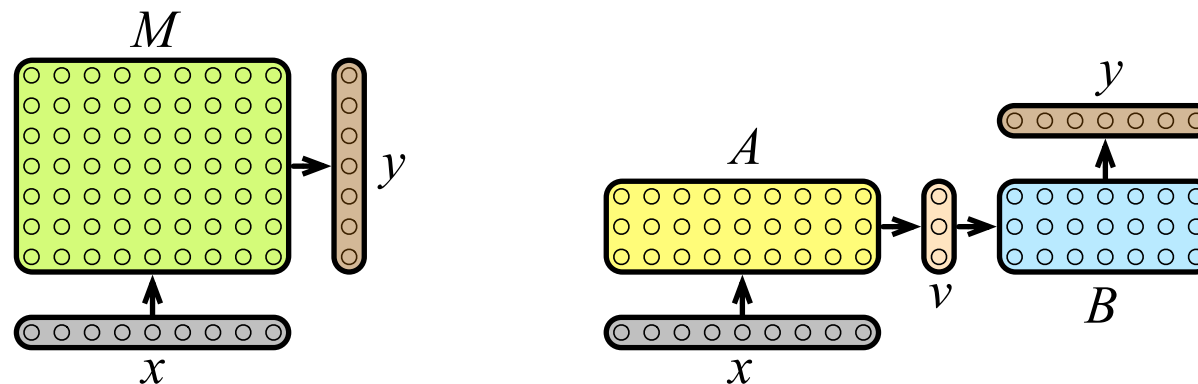
⇒ Need to reduce size of matrix

Factored Decomposition

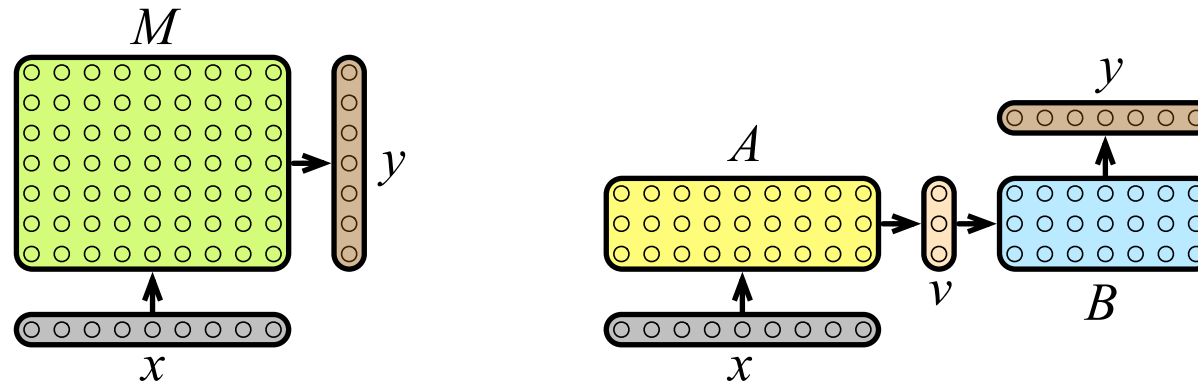
- One challenge: very large input and output vectors
- Number of parameters in matrix $M = |x| \times |y|$

⇒ Need to reduce size of matrix

- Solution: first reduce to smaller representation



Factored Decomposition: Math



- Intuition

- given highly dimension vector x
- first map to into lower dimensional vector v (matrix A)
- then map to output vector y (matrix B)

$$v = Ax$$

$$y = Bv = BAx$$

- Example

- $|x| = 20,000, |y| = 50,000 \rightarrow M = 1,000,000,000$
- $|v| = 100 \rightarrow A = 20,000 \times 100 = 2,000,000, B = 100 \times 50,000 = 5,000,000$
- reduction from $1,000,000,000$ to $7,000,000$

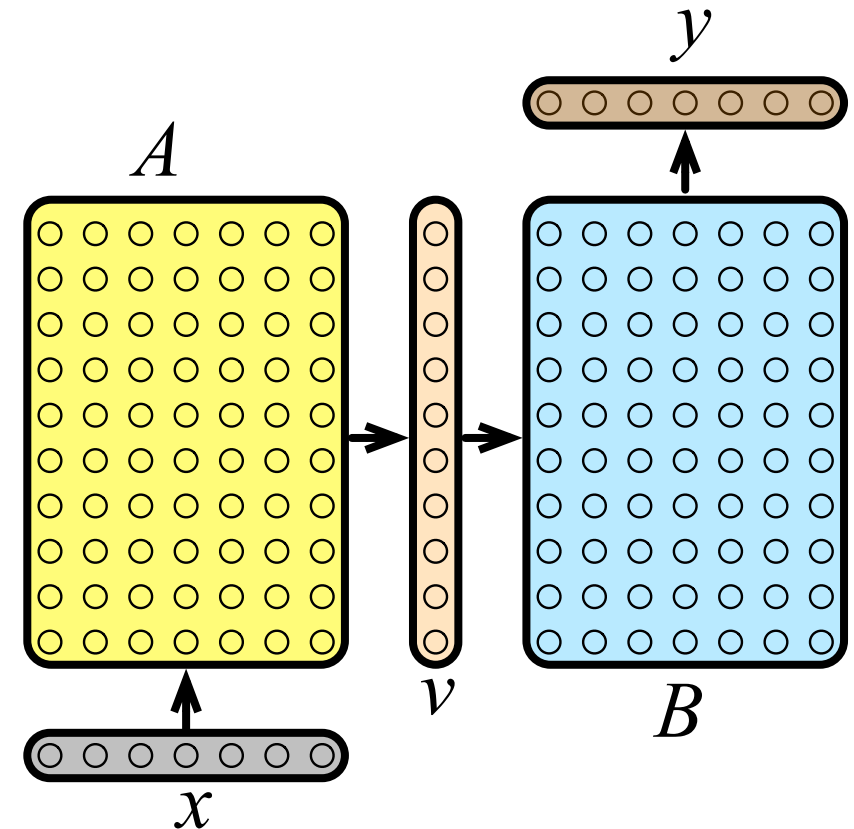
Factored Decomposition: Interpretation



- Vector v is a bottleneck feature
- Forced to capture salient features
- One example: word embeddings

Large Intermediate Vector

- Now for the opposite idea:
Large intermediate vector
- Allows for more complex
feature interactions
- But still: no increased
vector size at the end
- Example: $512 \rightarrow 2048 \rightarrow 512$
2 matrices with 1 million parameters



basic mathematical operations

Concatenation

- Often multiple input vectors to processing step
- For instance recurrent neural network
 - input word
 - previous state
- Combined in feed-forward layer

$$y = \text{activation}(M_1x_1 + M_2x_2 + b)$$

Concatenation

- Often multiple input vectors to processing step
- For instance recurrent neural network
 - input word
 - previous state
- Combined in feed-forward layer

$$y = \text{activation}(M_1x_1 + M_2x_2 + b)$$

- Another view

$$x = \text{concat}(x_1, x_2)$$

$$y = \text{activation}(Mx + b)$$

- Splitting hairs here, but concatenation useful generally

Addition

- Adding vectors: very simplistic, but often done
- Example: compute sentence embeddings s from word embeddings $w_1, \dots, w_n$

$$s = \sum_i^n w_i$$

- Reduces varying length sentence representation into fixed sized vector

- Adding vectors: very simplistic, but often done
- Example: compute sentence embeddings s from word embeddings $w_1, \dots, w_n$

$$s = \sum_i^n w_i$$

- Reduces varying length sentence representation into fixed sized vector
- Maybe weight the words, e.g., by attention

Multiplication

- Another elementary mathematical operation
- Three ways to multiply vectors

Multiplication

- Another elementary mathematical operation
- Three ways to multiply vectors
 - element-wise multiplication

$$v \odot u = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \odot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} v_1 \times u_1 \\ v_2 \times u_2 \end{pmatrix}$$

Multiplication

- Another elementary mathematical operation
- Three ways to multiply vectors
 - element-wise multiplication

$$v \odot u = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \odot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} v_1 \times u_1 \\ v_2 \times u_2 \end{pmatrix}$$

- dot product

$$v \cdot u = v^T u = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}^T \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = v_1 \times u_1 + v_2 \times u_2$$

used for simple version of attention mechanism

Multiplication

- Another elementary mathematical operation
- Three ways to multiply vectors
 - element-wise multiplication

$$v \odot u = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \odot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} v_1 \times u_1 \\ v_2 \times u_2 \end{pmatrix}$$

- dot product

$$v \cdot u = v^T u = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}^T \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = v_1 \times u_1 + v_2 \times u_2$$

used for simple version of attention mechanism

- third possibility: vu^T , not commonly done

Maximum

- Goal: reduce the dimensionality of representation
- Example: detect if a face is in image
 - any region of image may have positive match
 - represent different regions with element in a vector
 - maximum value: any region has a face

- Goal: reduce the dimensionality of representation
- Example: detect if a face is in image
 - any region of image may have positive match
 - represent different regions with element in a vector
 - maximum value: any region has a face
- Max pooling
 - given: n dimensional vector
 - goal: reduce to $\frac{n}{k}$ dimensional vector
 - method: break up vector into blocks of k elements, map each into single value

Max Out

- Max out
 - first branch out into multiple feed-forward layers

$$W_1x + b_1$$

$$W_2x + b_2$$

Max Out

- Max out
 - first branch out into multiple feed-forward layers

$$W_1x + b_1$$

$$W_2x + b_2$$

- element-wise maximum

$$\text{maxout}(x) = \max(W_1x + b_1, W_2x + b_2)$$

- Max out
 - first branch out into multiple feed-forward layers

$$W_1x + b_1$$

$$W_2x + b_2$$

- element-wise maximum

$$\text{maxout}(x) = \max(W_1x + b_1, W_2x + b_2)$$

- ReLu activation is a maxout layer: maximum of feed-forward layer and 0

$$\text{ReLU}(x) = \max(Wx + b, 0)$$



processing sequences

Recurrent Neural Networks

- Already described recurrent neural networks at length
 - propagate state s
 - over time steps t
 - receiving an input x_t at each turn

$$s_t = f(s_{t-1}, x_t)$$

(state may computed may as a feed-forward layer)

Recurrent Neural Networks

- Already described recurrent neural networks at length
 - propagate state s
 - over time steps t
 - receiving an input x_t at each turn

$$s_t = f(s_{t-1}, x_t)$$

(state may computed may as a feed-forward layer)

- More successful
 - gated recurrent units (GRU)
 - long short-term memory cells (LSTM)

- Already described recurrent neural networks at length
 - propagate state s
 - over time steps t
 - receiving an input x_t at each turn

$$s_t = f(s_{t-1}, x_t)$$

(state may computed may as a feed-forward layer)

- More successful
 - gated recurrent units (GRU)
 - long short-term memory cells (LSTM)
- Good fit for sequences, like words in a sentence
 - humans also receive word by word
 - most recent words most relevant
 - closer to current state
- But computational problematic: very long computation chains

Alternative Sequence Processing

- Convolutional neural networks
- Attention

convolutional neural networks

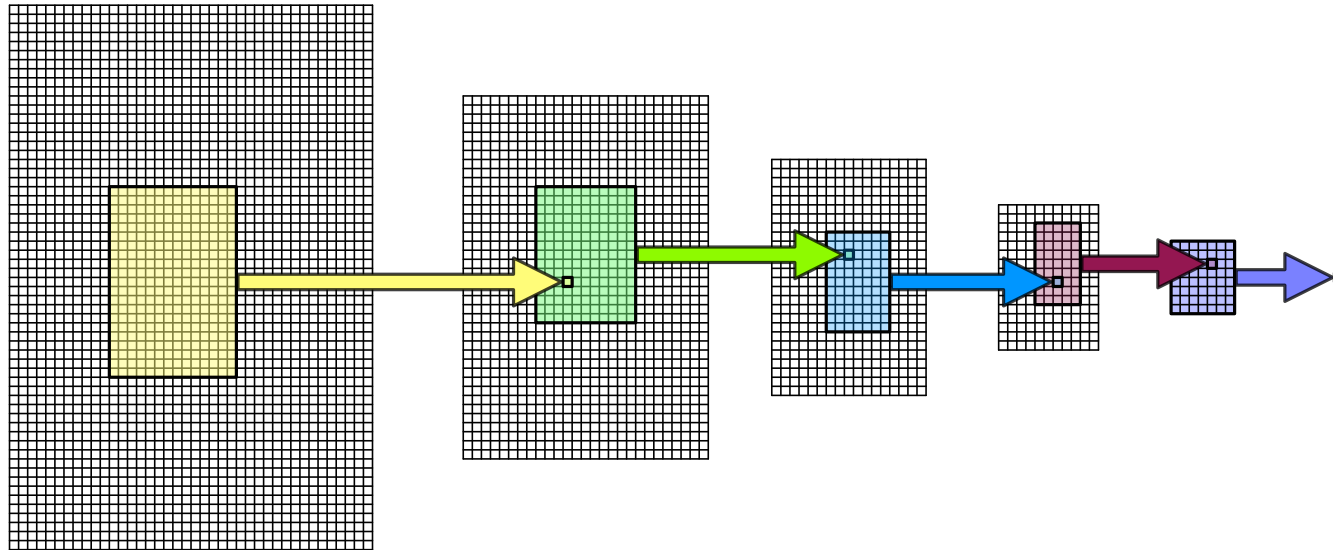
Convolutional Neural Networks

- Key step
 - take a high dimensional input representation
 - map to lower dimensional representation
- Several repetitions of this step

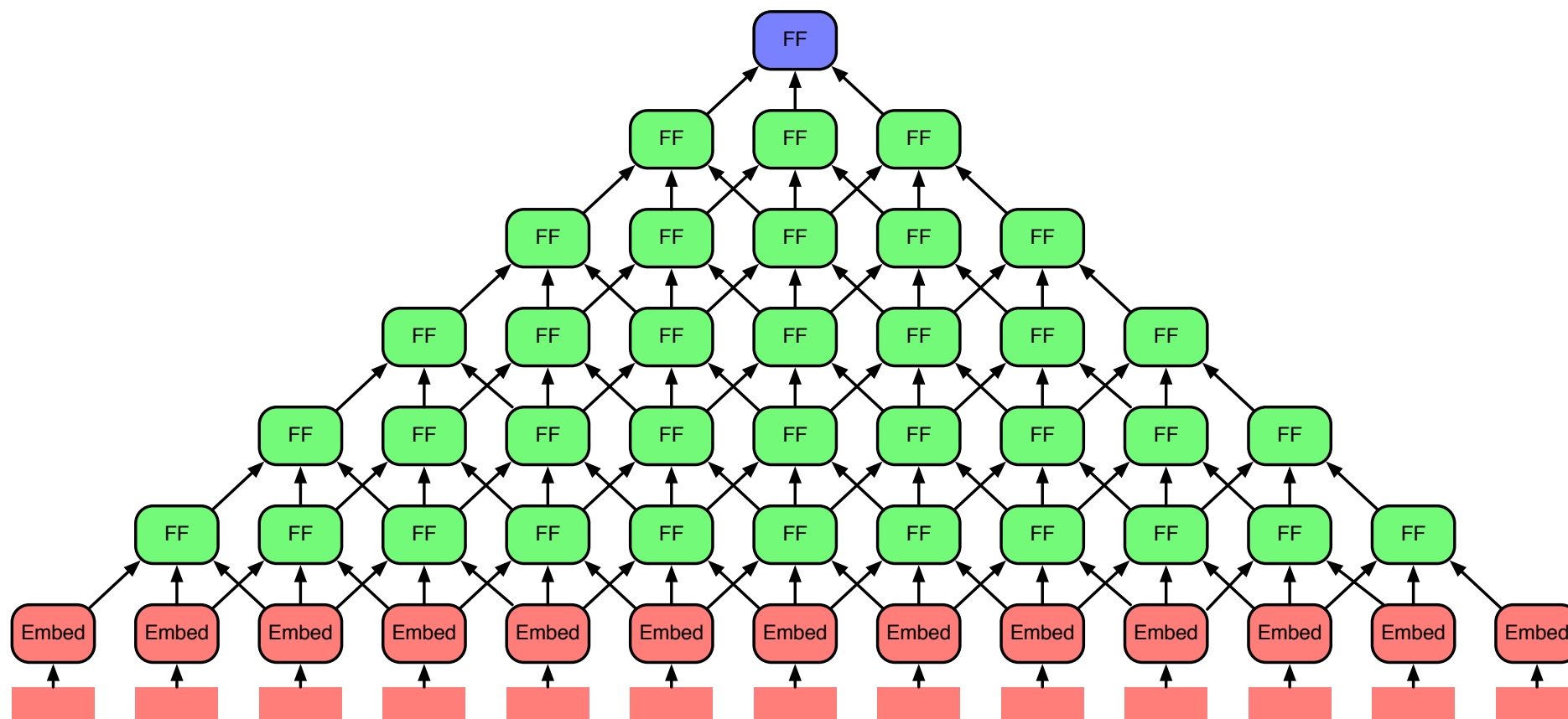
Convolutional Neural Networks

- Key step
 - take a high dimensional input representation
 - map to lower dimensional representation
- Several repetitions of this step
- Examples
 - map 50×50 pixel area into scalar value
 - combine 3 or more neighboring words into a single vector

- Key step
 - take a high dimensional input representation
 - map to lower dimensional representation
- Several repetitions of this step
- Examples
 - map 50×50 pixel area into scalar value
 - combine 3 or more neighboring words into a single vector
- Machine translation
 - encode input sentence into single vector
 - decode this vector into a sentence in the output language



- Popular in image processing
- Regions of an image are reduced into increasingly smaller representation
 - matrix spanning part of image reduced to single value
 - overlapping regions



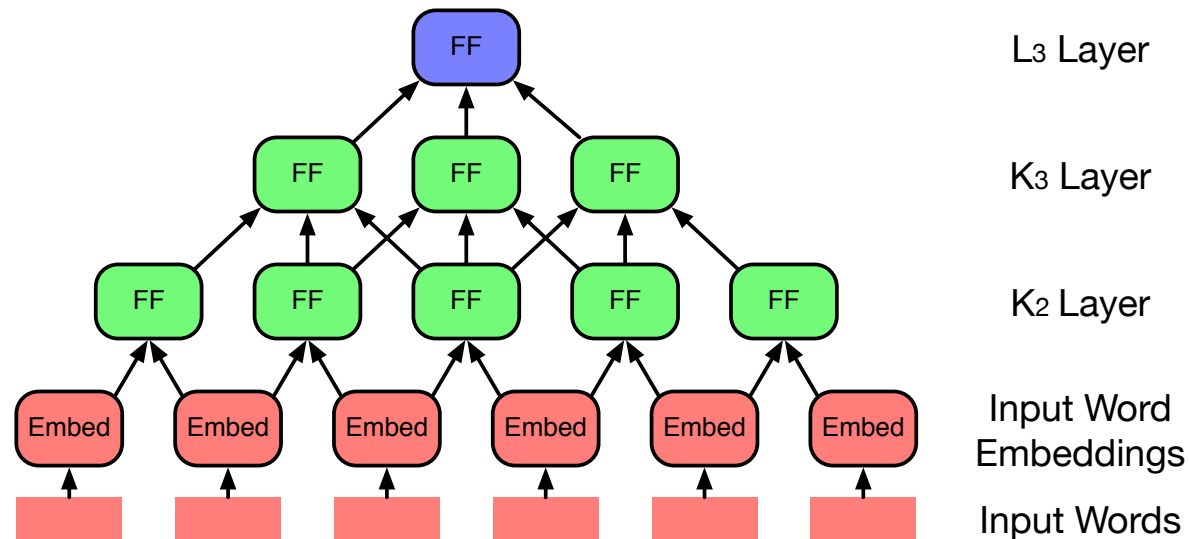
- Map words into fixed-sized sentence representation

Hierarchical Structure and Language

- Syntactic and semantic theories of language
 - language is recursive
 - central: verb
 - dependents: subject, objects, adjuncts
 - their dependents: adjectives, determiners
 - also nested: relative clauses
- How to compute sentence embeddings active research topic

Convolutional Machine Translation

- First end-to-end neural machine translation model of the modern era
[Kalchbrenner and Blunsom, 2013]
- Encoder



- always two convolutional layers, with different size
 - here: K_2 and K_3
- Decoder similar

attention

Attention

- Machine translation is a structured prediction task
 - output is not a single label
 - output structure needs to be built, word by word

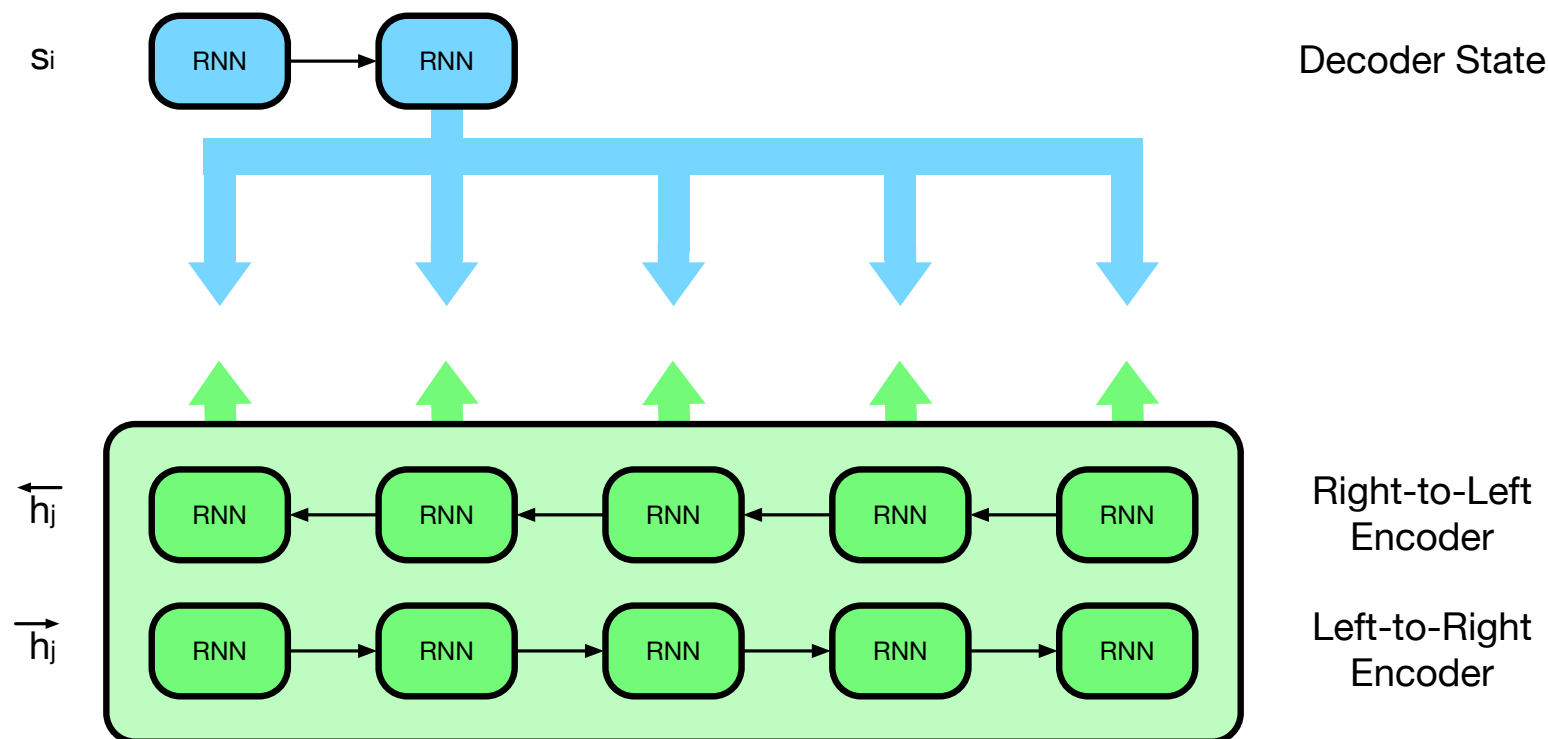
Attention

- Machine translation is a structured prediction task
 - output is not a single label
 - output structure needs to be built, word by word
- Relevant information for each word prediction varies

- Machine translation is a structured prediction task
 - output is not a single label
 - output structure needs to be built, word by word
- Relevant information for each word prediction varies
- Human translators pay attention to different parts of the input sentence when translating

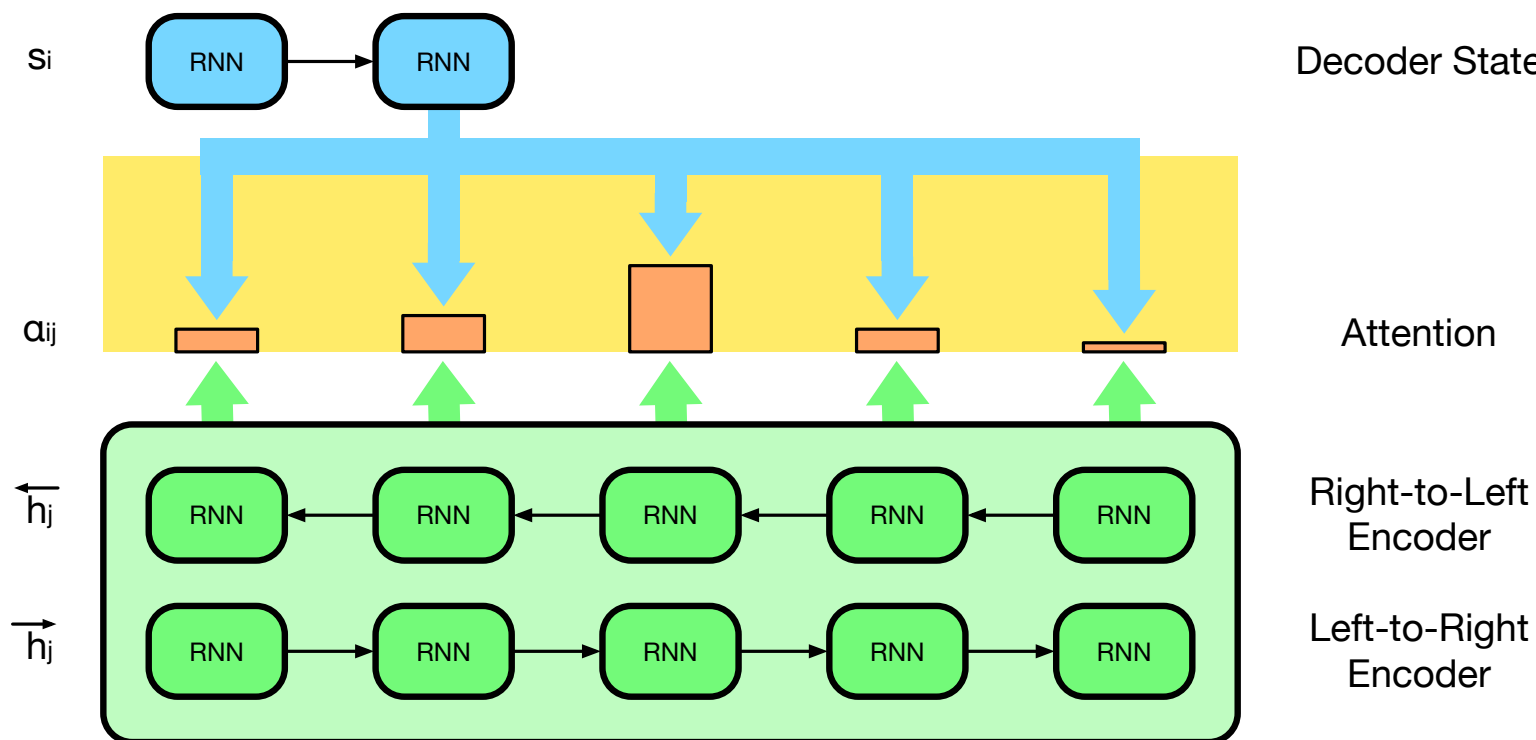
⇒ Attention mechanism

Attention



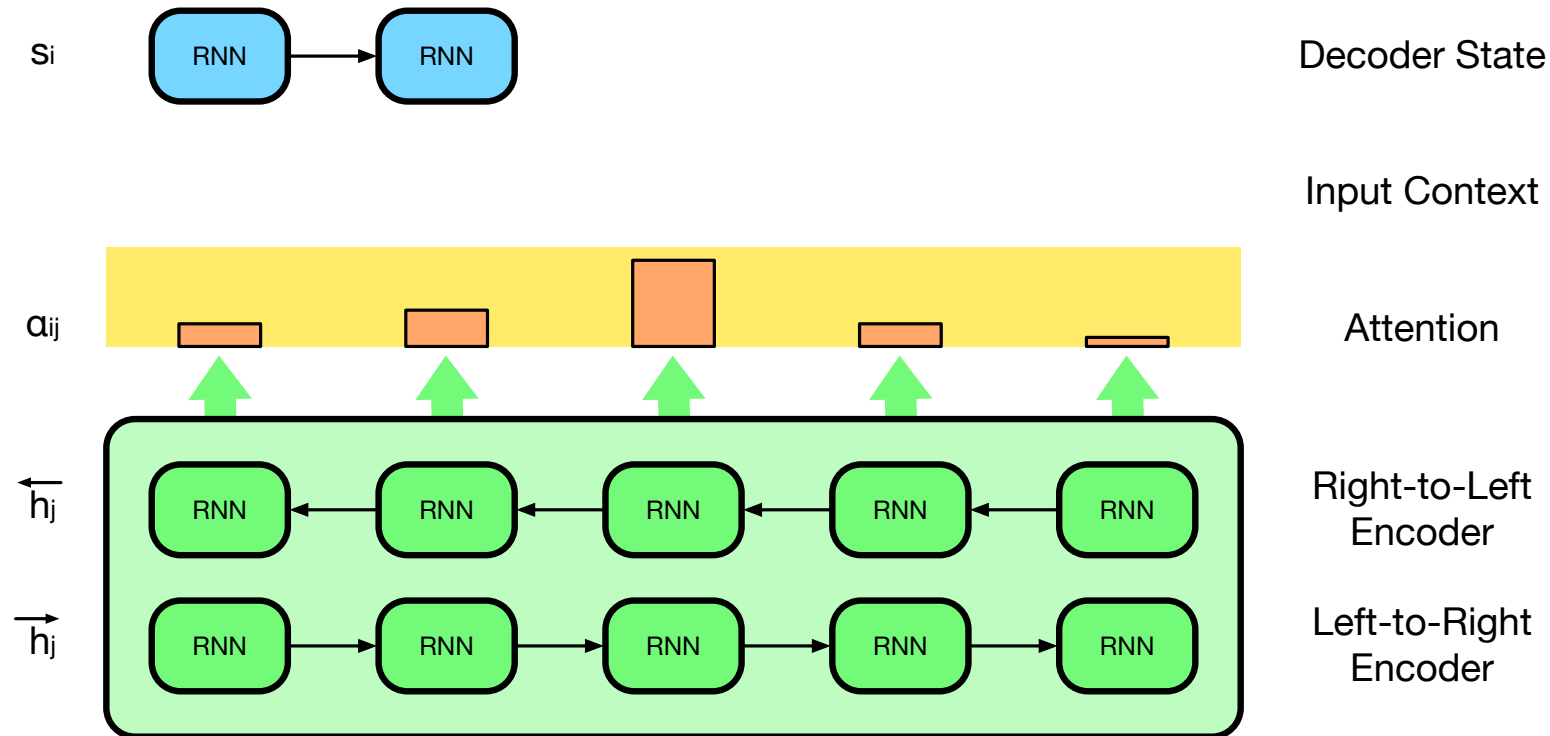
- Given what we have generated so far (decoder hidden state)
- ... which words in the input should we pay attention to (encoder states)?

Attention



- Given: – the previous hidden state of the decoder s_{i-1}
– the representation of input words $h_j = (\overleftarrow{h_j}, \overrightarrow{h_j})$
- Predict an alignment probability $a(s_{i-1}, h_j)$ to each input word j
(modeled with with a feed-forward neural network layer)

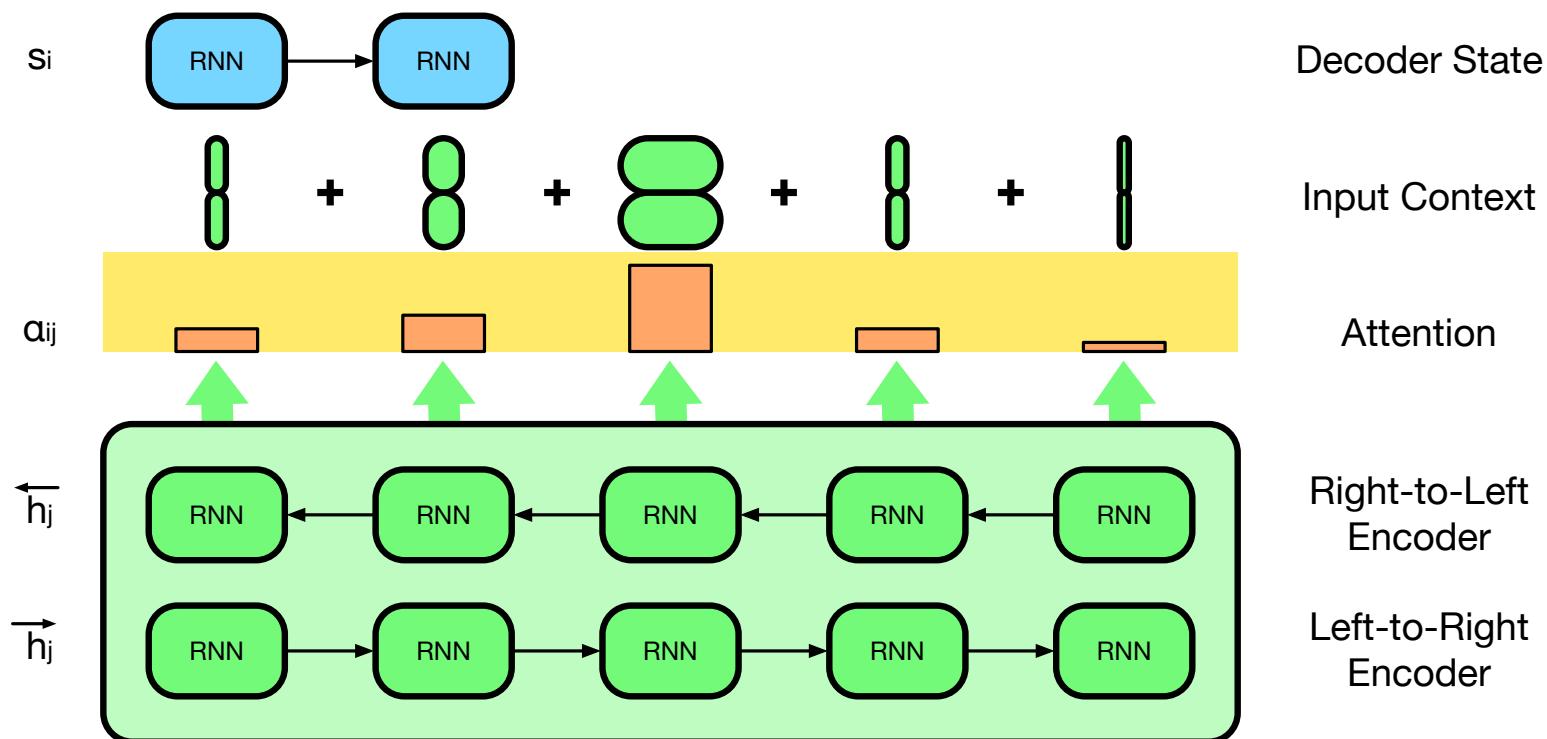
Attention



- Normalize attention (softmax)

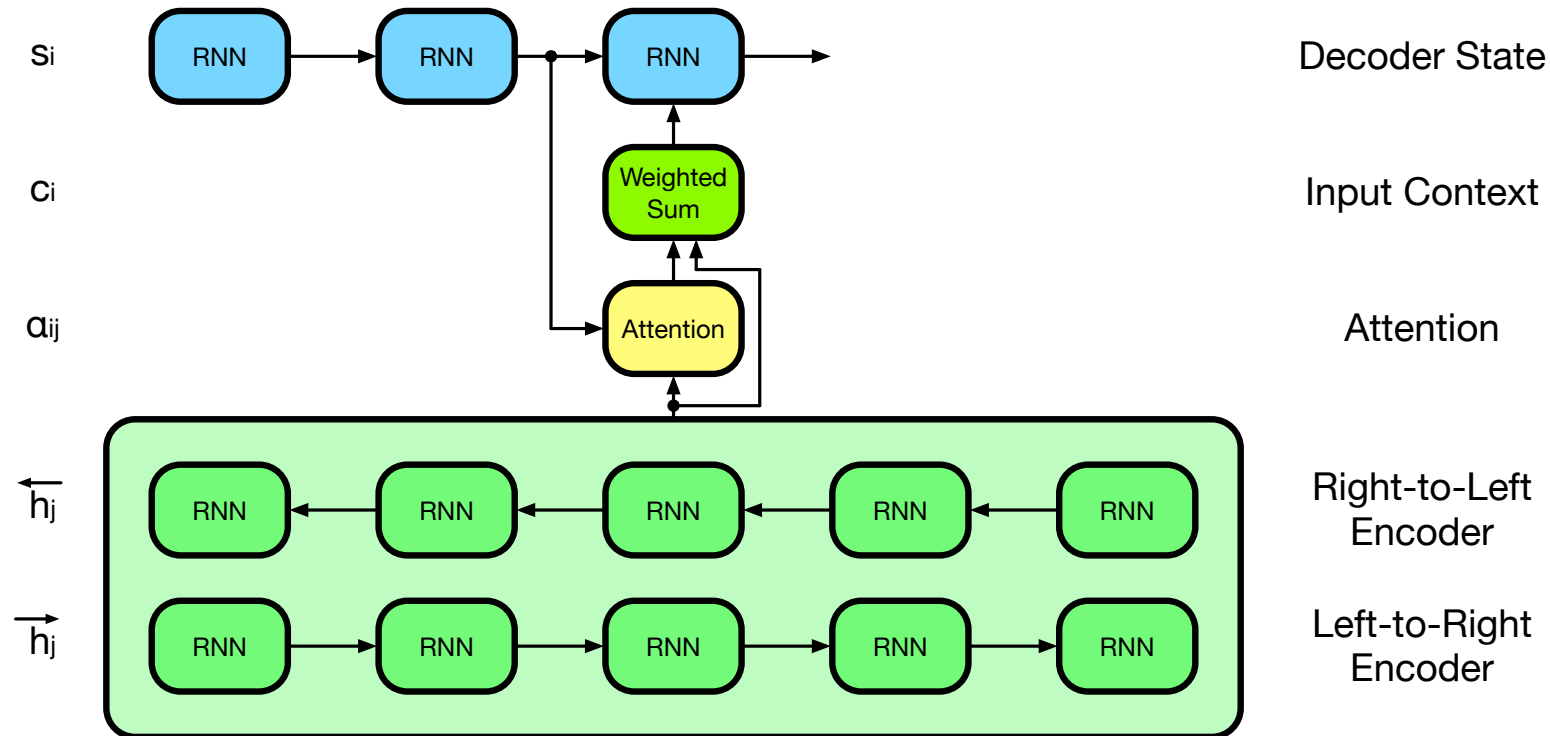
$$\alpha_{ij} = \frac{\exp(a(s_{i-1}, h_j))}{\sum_k \exp(a(s_{i-1}, h_k))}$$

Attention



- Relevant input context: weigh input words according to attention: $c_i = \sum_j \alpha_{ij} h_j$

Attention



- Use context to predict next hidden state and output word

Computing Attention

- Attention mechanism in neural translation model (Bahdanau et al., 2015)
 - previous hidden state s_{i-1}
 - input word embedding h_j
 - trainable parameters b, W_a, U_a, v_a

$$a(s_{i-1}, h_j) = v_a^T \tanh(W_a s_{i-1} + U_a h_j + b)$$

- Attention mechanism in neural translation model (Bahdanau et al., 2015)
 - previous hidden state s_{i-1}
 - input word embedding h_j
 - trainable parameters b, W_a, U_a, v_a

$$a(s_{i-1}, h_j) = v_a^T \tanh(W_a s_{i-1} + U_a h_j + b)$$

- Other ways to compute attention (Luong et al., 2015)
 - Dot product: $a(s_{i-1}, h_j) = s_{i-1}^T h_j$
 - Scaled dot product: $a(s_{i-1}, h_j) = \frac{1}{\sqrt{|h_j|}} s_{i-1}^T h_j$
 - General: $a(s_{i-1}, h_j) = s_{i-1}^T W_a h_j$
 - Local: $a(s_{i-1}) = W_a s_{i-1}$

- Three elements

Query : decoder state

Key : encoder state

Value : encoder state

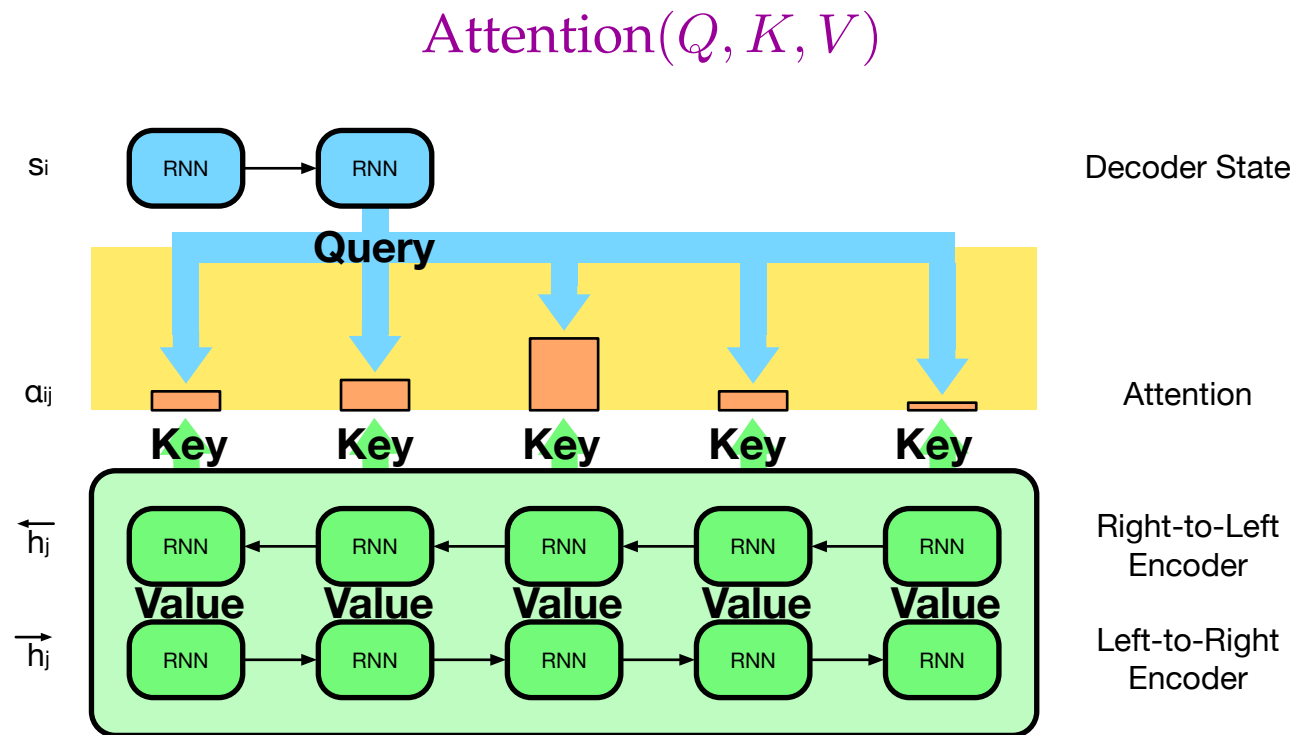
- Intuition

- given a query (the decoder state)
- we check how well it matches keys in the database (the encoder states)
- and then use the matching score to scale the retrieved value (also the encoder state)

- Computation

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

General View of Dot-Product Attention



- Query: encoder state, Key and Value: decoder state

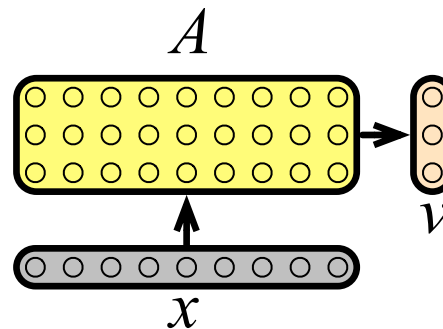
$$\text{Attention}(S, H, H)$$

Dimensionality Reduction

- Instead of simple dot product of query and key vectors QK^T ...
- Multiply with weight matrices W^Q and W^K

Dimensionality Reduction

- Instead of simple dot product of query and key vectors QK^T ...
- Multiply with weight matrices W^Q and W^K
- Also reduce the size of the vectors



- New computation: $\text{Attention}(QW^Q, KW^K, V)$

Self Attention

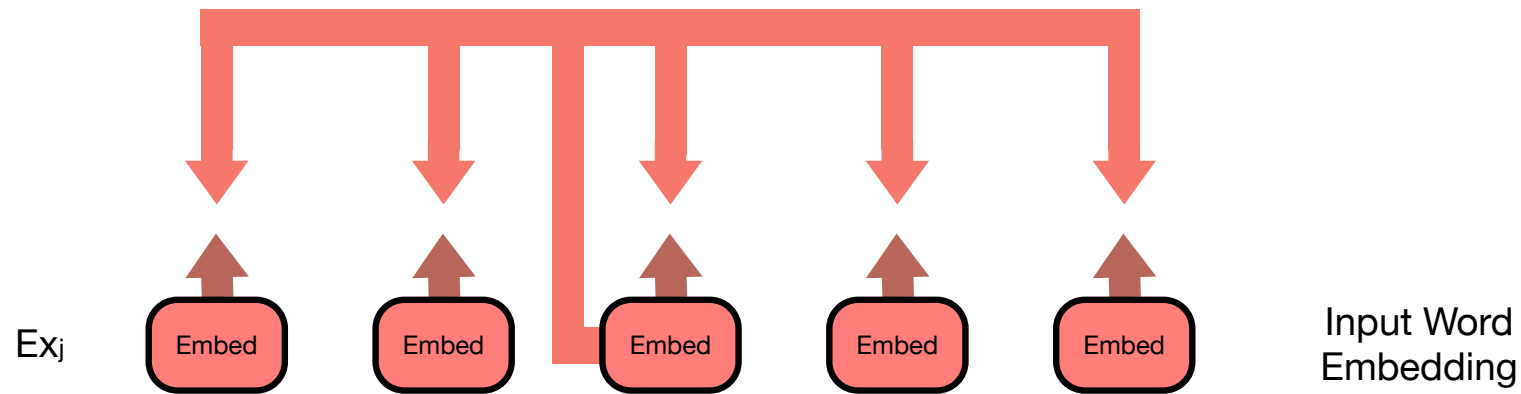
- Finally, a very different take at attention
- Motivation so far: need for alignment between input words and output words

Self Attention

- Finally, a very different take at attention
- Motivation so far: need for alignment between input words and output words
- Now: refine representation of input words in the encoder
 - representation of an input word mostly depends on itself
 - but also informed by the surrounding context
 - previously: recurrent neural networks (considers left or right context)
 - now: attention mechanism

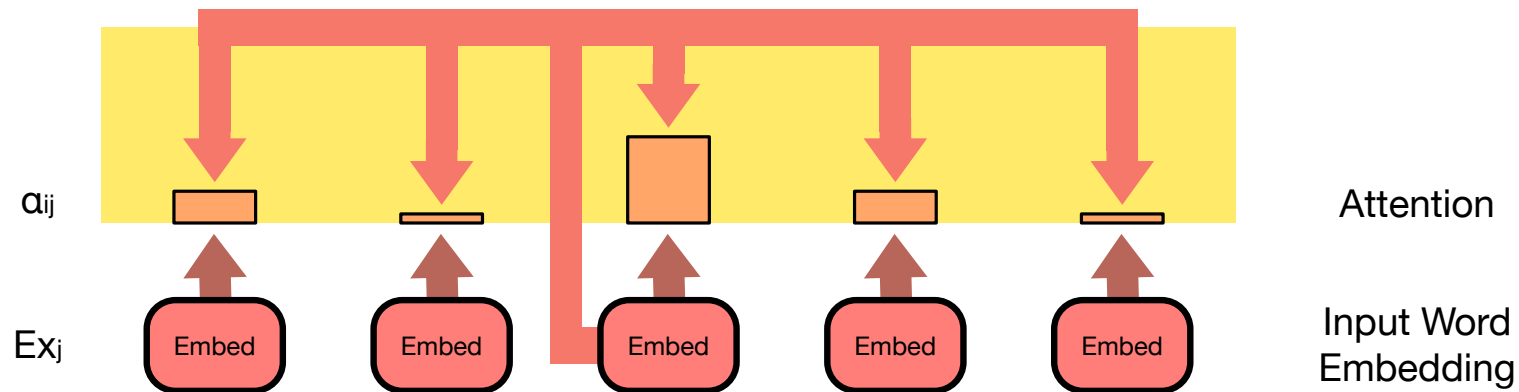
- Finally, a very different take at attention
- Motivation so far: need for alignment between input words and output words
- Now: refine representation of input words in the encoder
 - representation of an input word mostly depends on itself
 - but also informed by the surrounding context
 - previously: recurrent neural networks (considers left or right context)
 - now: attention mechanism
- Self attention:
Which of the surrounding words is most relevant to refine representation?

Self Attention



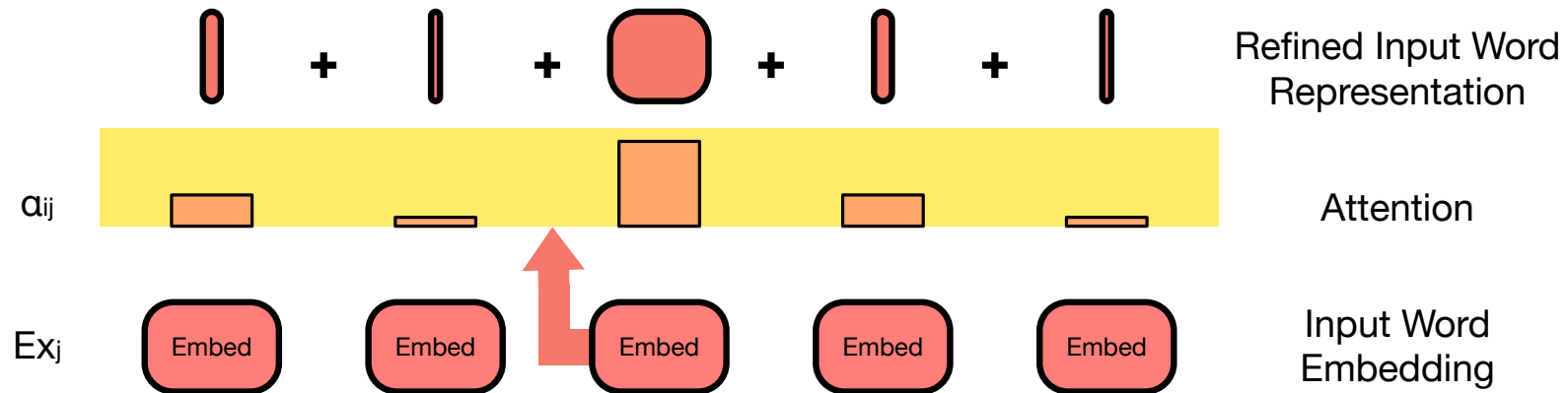
- Given: input word embeddings
- Task: consider how each should be refined in view of others
- Needed: how much attention to pay to others

Self Attention



- Computation of attention weights as before
 - Key: word embedding (or generally: encoder state for word H)
 - Query: word embedding (or generally: encoder state for word H)
- Again, multiple with weight matrices: $Q=HW^Q$ and $K=HW^K$
- Attention weights: QK^T

Self Attention



- Full self attention

$$\text{self-attention}(H) = \text{Attention}(HW^Q, HW^K, H)$$

- Resulting vector uses weighted context words

Multi-Head Attention

- Add redundancy
 - say, 16 attention weights
 - each based on its own parameters W_i^Q, W_i^K, W_i^V

Multi-Head Attention

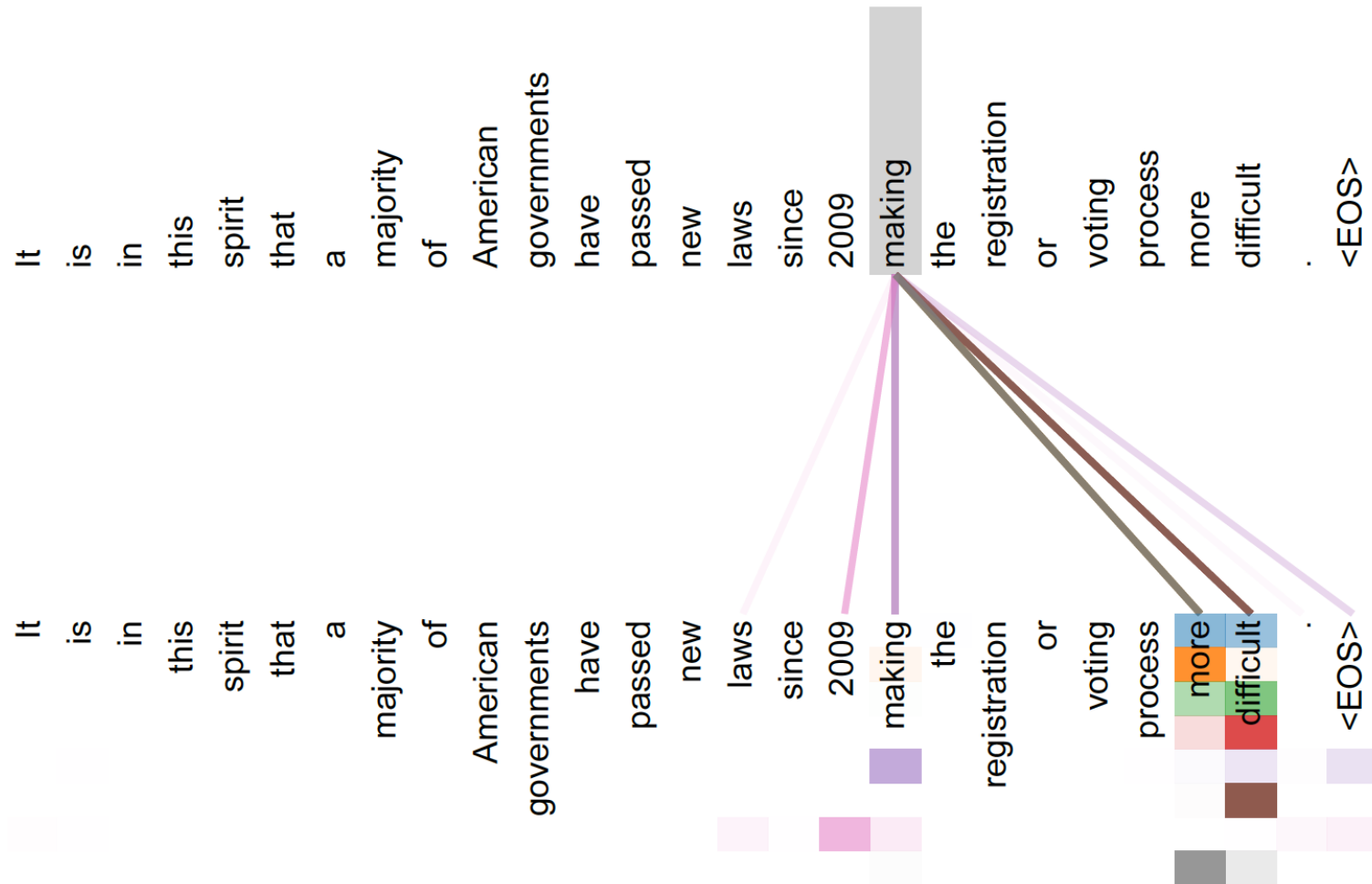
- Add redundancy
 - say, 16 attention weights
 - each based on its own parameters W_i^Q, W_i^K, W_i^V

- Formally:

$$\begin{aligned}\text{head}_i &= \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \\ \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O\end{aligned}$$

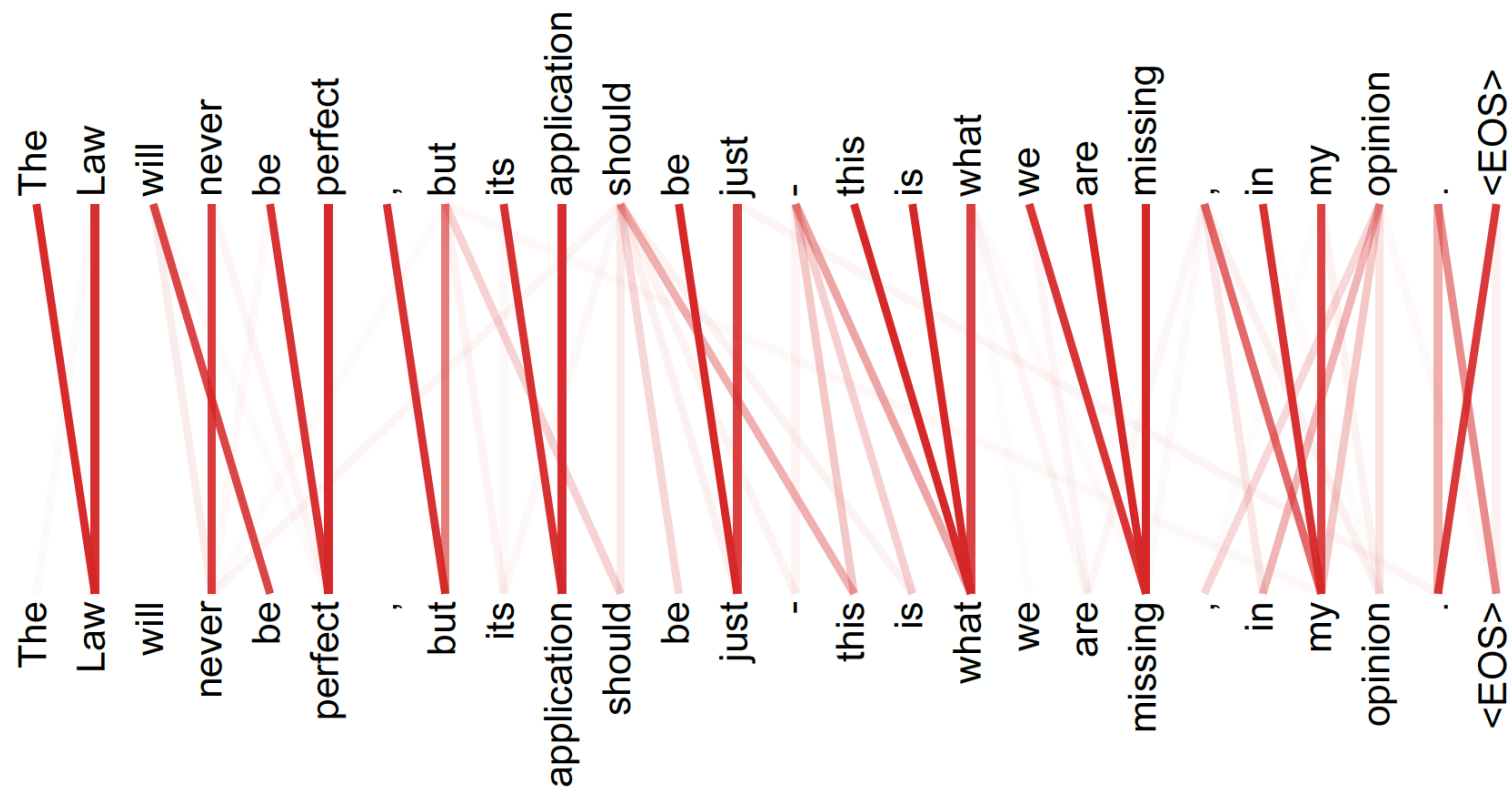
- Multi-head attention is a form of ensembling

Multi-Head Attention



Multi-Head Attention

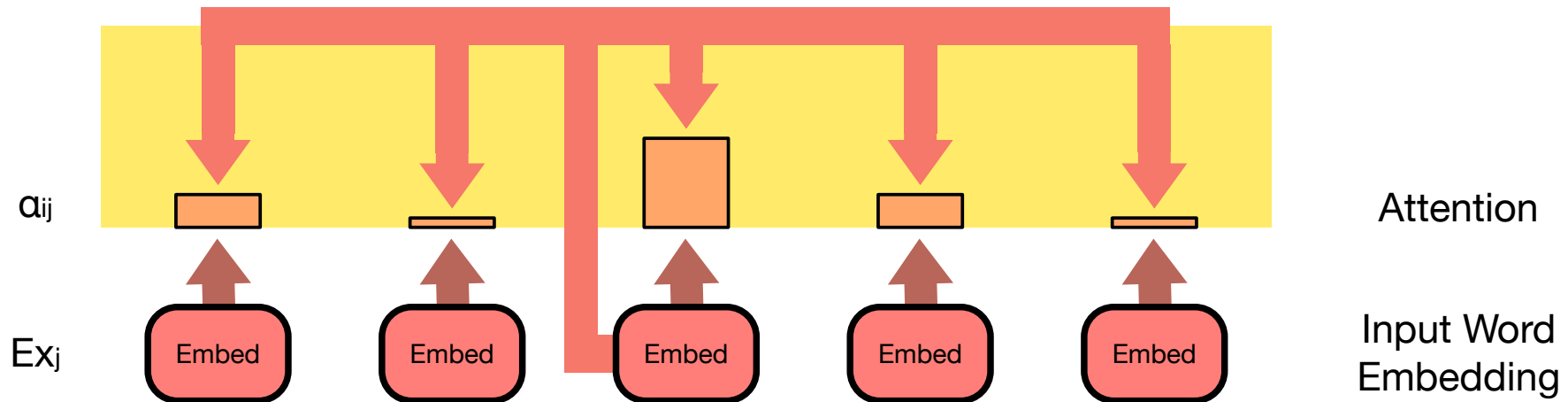
42



“Many of the attention heads exhibit behaviour that seems related to the structure of the sentence.”

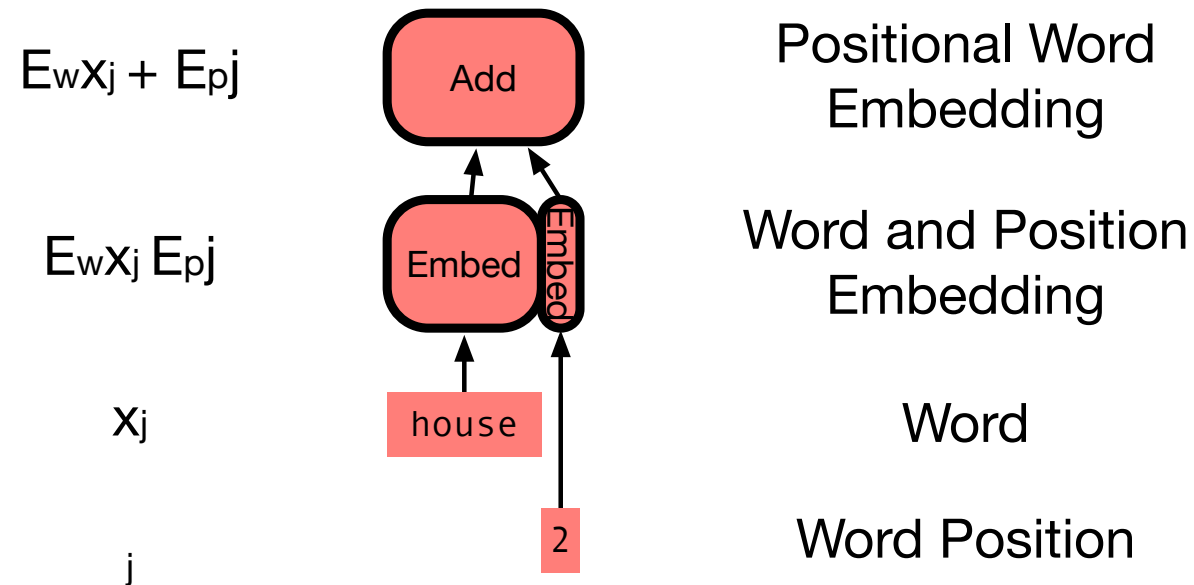
positional embeddings

Word Order?



- When computing attention weights
 - only consider word embeddings
 - not position of word in sentence
- But word order should matter...

Positional Embedding



- Add word position to the word embedding

Positional Embedding

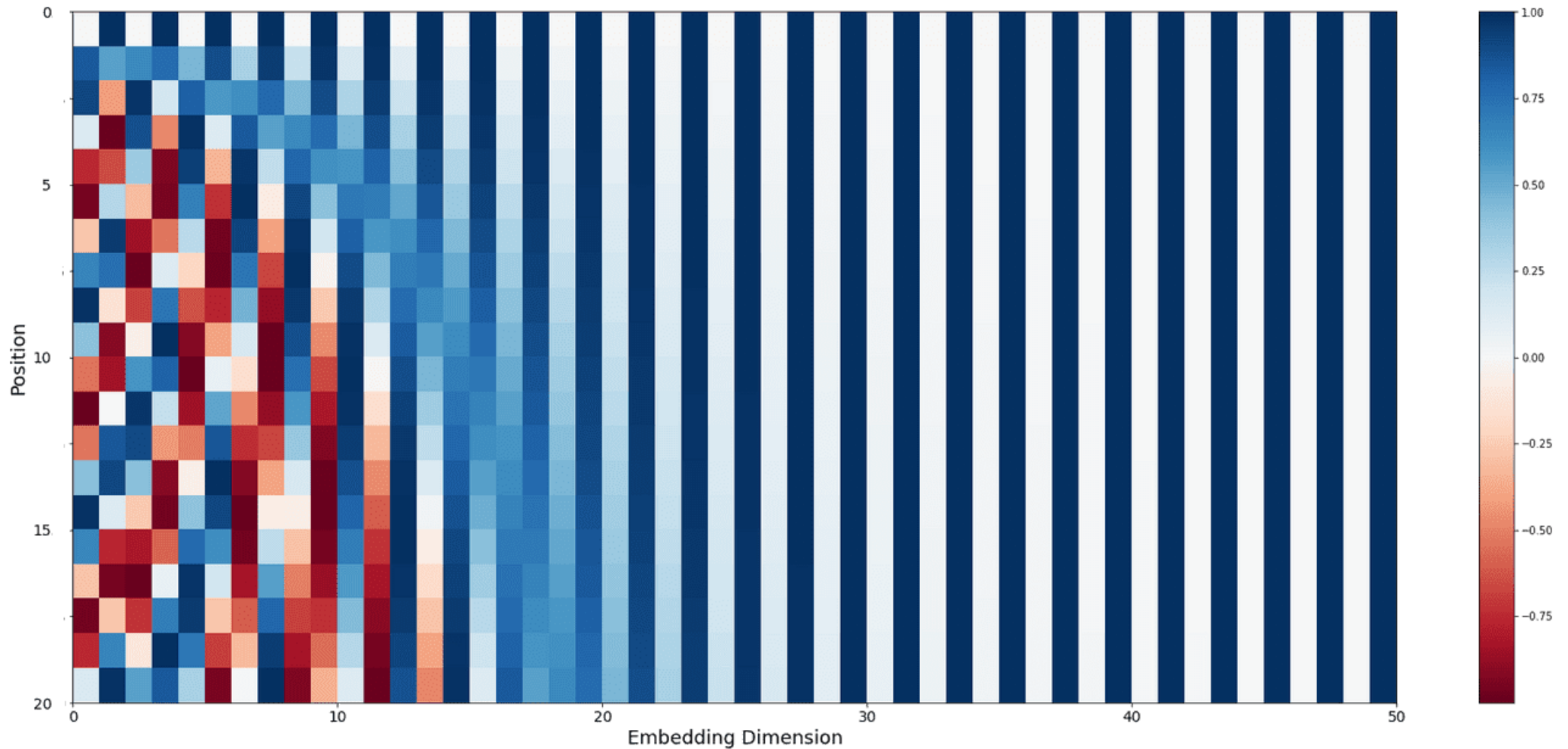
Define embedding with sine and cosine functions of different frequencies

$$\begin{aligned}\text{PE}(\textit{pos}, 2i) &= \sin\left(\frac{\textit{pos}}{10000^{2i/d}}\right) \\ \text{PE}(\textit{pos}, 2i + 1) &= \cos\left(\frac{\textit{pos}}{10000^{2i/d}}\right)\end{aligned}$$

i = index in embedding vector

d = dimension of embedding vector

Positional Embedding



transformer

Self Attention: Transformer

- Self-attention in encoder
 - refine word representation based on relevant context words
 - relevance determined by self attention

Self Attention: Transformer

- Self-attention in encoder
 - refine word representation based on relevant context words
 - relevance determined by self attention
- Self-attention in decoder
 - refine output word predictions based on relevant previous output words
 - relevance determined by self attention

Self Attention: Transformer

- Self-attention in encoder
 - refine word representation based on relevant context words
 - relevance determined by self attention
- Self-attention in decoder
 - refine output word predictions based on relevant previous output words
 - relevance determined by self attention
- Also regular attention to encoder states in decoder

- Self-attention in encoder
 - refine word representation based on relevant context words
 - relevance determined by self attention
- Self-attention in decoder
 - refine output word predictions based on relevant previous output words
 - relevance determined by self attention
- Also regular attention to encoder states in decoder
- Currently most successful model
(maybe only with self attention in decoder, but regular recurrent decoder)

Self Attention Layer

- Given: input word representations h_j , packed into a matrix $H = (h_1, \dots, h_j)$
- Self attention $\text{self-attention}(H) = \text{MultiHead}(H, H, H)$

Self Attention Layer

- Given: input word representations h_j , packed into a matrix $H = (h_1, \dots, h_j)$

- Self attention $\text{self-attention}(H) = \text{MultiHead}(H, H, H)$

- Shortcut connection $\text{self-attention}(h_j) + h_j$

Self Attention Layer

- Given: input word representations h_j , packed into a matrix $H = (h_1, \dots, h_j)$
- Self attention $\text{self-attention}(H) = \text{MultiHead}(H, H, H)$
- Shortcut connection $\text{self-attention}(h_j) + h_j$
- Layer normalization

Self Attention Layer

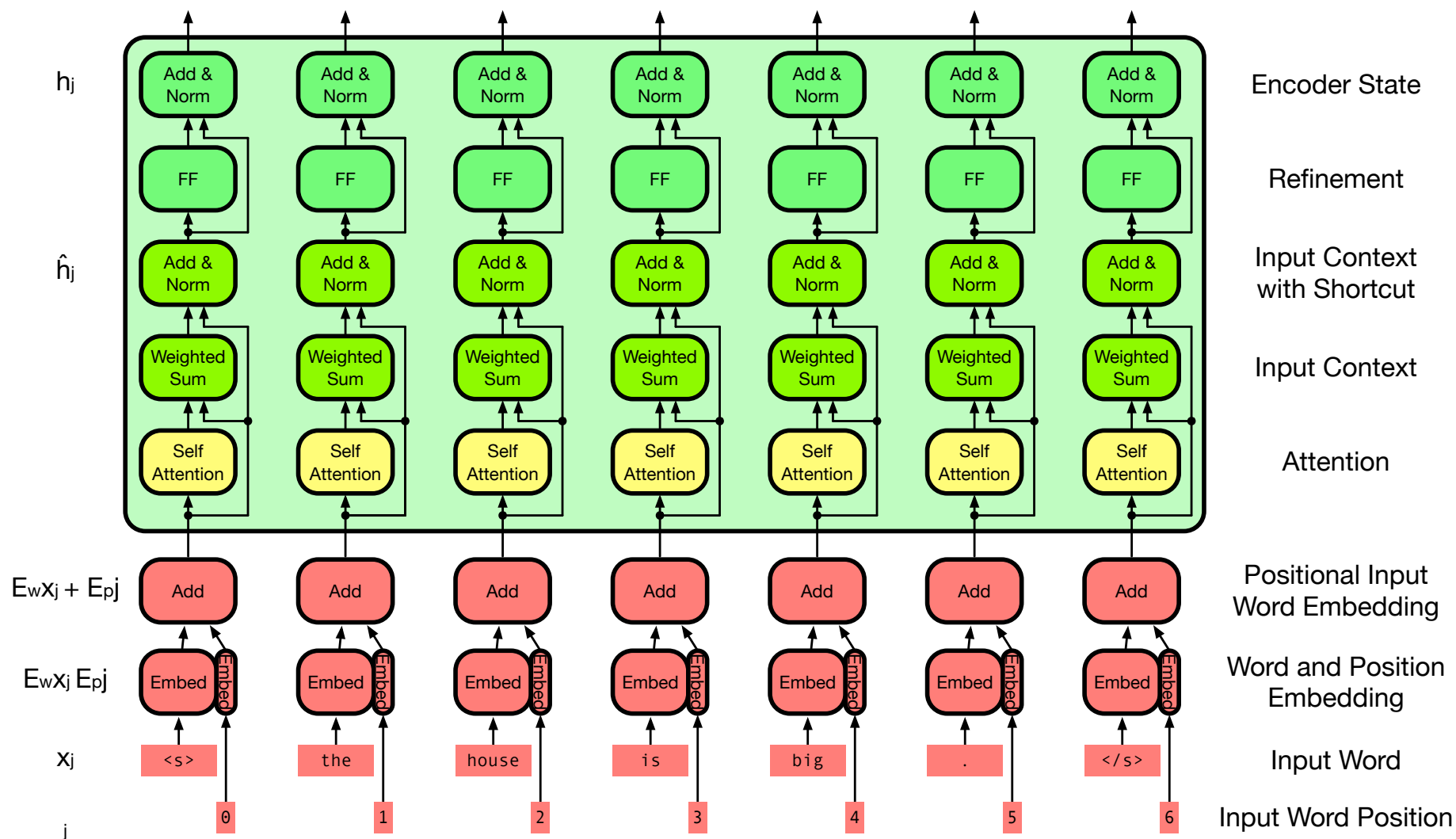
- Given: input word representations h_j , packed into a matrix $H = (h_1, \dots, h_j)$
- Self attention $\text{self-attention}(H) = \text{MultiHead}(H, H, H)$
- Shortcut connection $\text{self-attention}(h_j) + h_j$
- Layer normalization
- Feed-forward step with ReLU activation function and large intermediate vector

$$\text{relu}(W_1 \hat{h}_j + b_1) W_2 + b_2$$

Self Attention Layer

- Given: input word representations h_j , packed into a matrix $H = (h_1, \dots, h_j)$
- Self attention
 $\text{self-attention}(H) = \text{MultiHead}(H, H, H)$
- Shortcut connection
 $\text{self-attention}(h_j) + h_j$
- Layer normalization
- Feed-forward step with ReLU activation function and large intermediate vector
 $\text{relu}(W_1 \hat{h}_j + b_1)W_2 + b_2$
- Again, shortcut connection and layer normalization

Encoder



Sequence of self-attention layers

Self-Attention in the Decoder

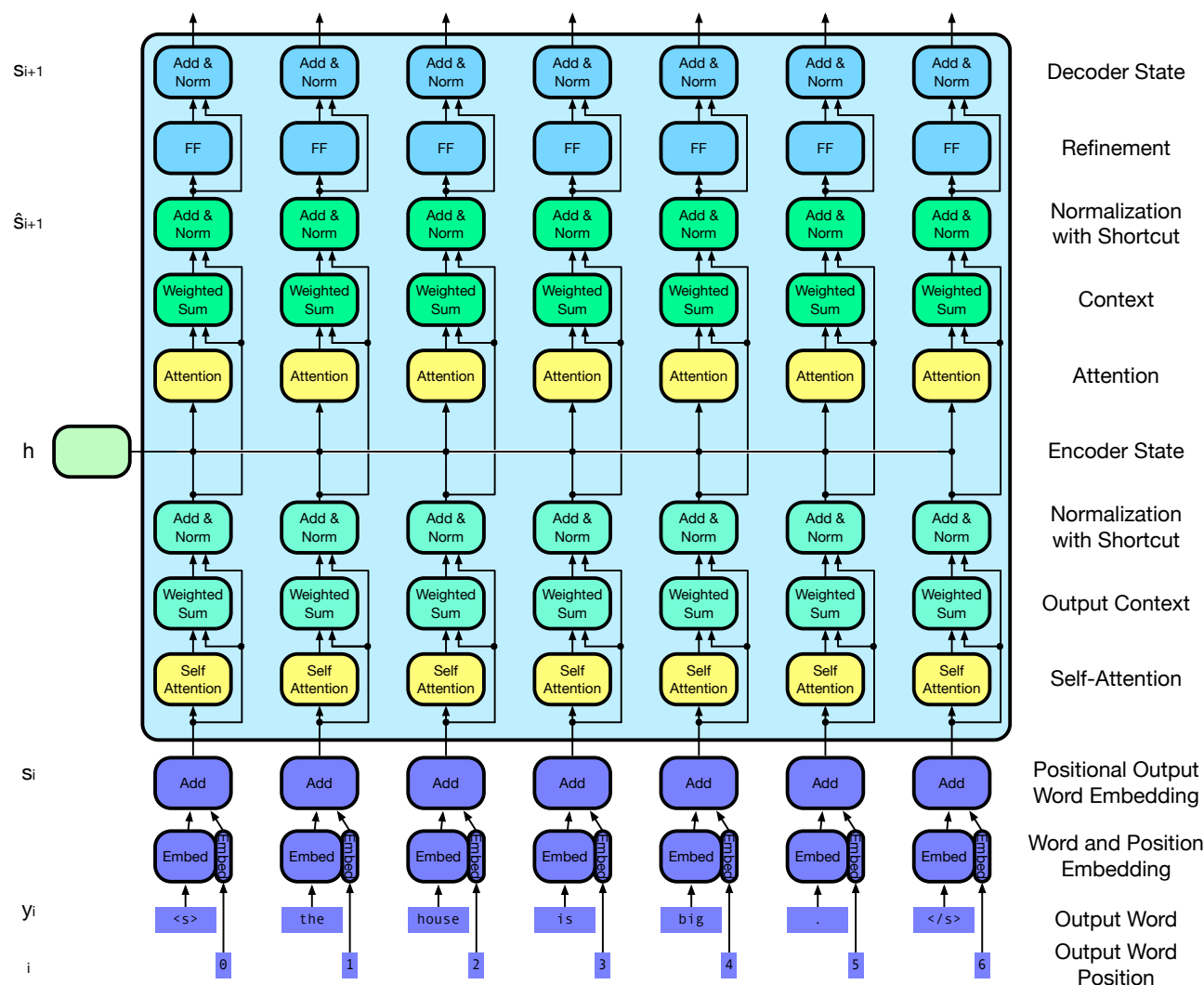
- Same idea as in the encoder
- Output words are initially encoded by word embeddings $s_i = Ey_i$.
- Self attention is computed over previous output words
 - association of a word s_i is limited to words s_k ($k \leq i$)
 - resulting representation $\tilde{s}_i$

$$\text{self-attention}(\tilde{S}) = \text{MultiHead}(\tilde{S}, \tilde{S}, \tilde{S})$$

- Original intuition of attention mechanism: focus on relevant input words
- Computed with dot product $\tilde{S}H^T$
- Compute attention between the decoder states $\tilde{S}$ and the final encoder states H
$$\text{attention}(\tilde{S}, H) = \text{MultiHead}(\tilde{S}, H, H)$$
- Note: attention mechanism formally mirrors self-attention

- Self-attention $\text{self-attention}(\tilde{S}) = \text{MultiHead}(\tilde{S}, \tilde{S}, \tilde{S})$
 - shortcut connections
 - layer normalization
- Attention $\text{attention}(\tilde{S}, H) = \text{softmaxMultiHead}(\tilde{S}, H, H)$
 - shortcut connections
 - layer normalization
 - feed-forward layers with large intermediate vector
- Multiple stacked layers

Decoder



Decoder computes attention-based representations of the output in several layers, initialized with the embeddings of the previous output words

Multiple Layers

- Stack several transformer layers (say, $D = 6$)

- Encoder

- Start with input word embedding

$$h_{0,j} = Ex_j$$

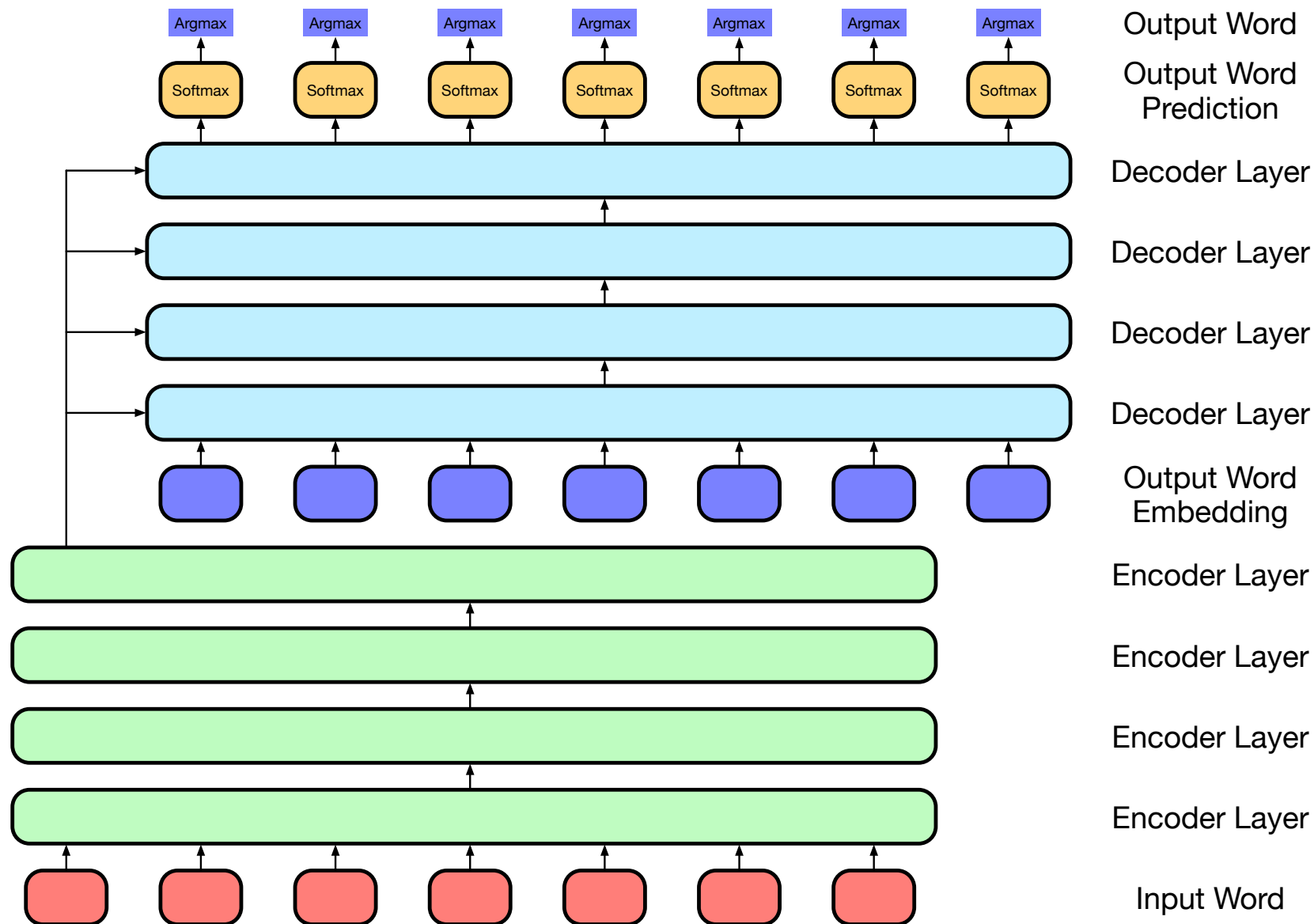
- Stacked layers

$$h_{d,j} = \text{self-attention-layer}(h_{d-1,j})$$

- Same for decoder

Multiple Layers in Encoder and Decoder

57



Model Sizes: Transformer Big

- 6 encoder and 6 decoder layers
- Vector size 1024
- Vector inbetween the two feed-forward layers: 4096
- Attention heads: 16
- Scaled down vector for key/query computation: 64

questions?